



Introduction

Neuromorphic computing replicates brain structure and function directly in hardware. Unlike conventional processors, it uses parallel, event-driven processing for greater efficiency. Software-based spiking neural networks (SNN) are a big player in autonomous control systems, but they carry high energy and resource costs.

SNN 1 input 4 neurons and 1 DOF pong paddle control

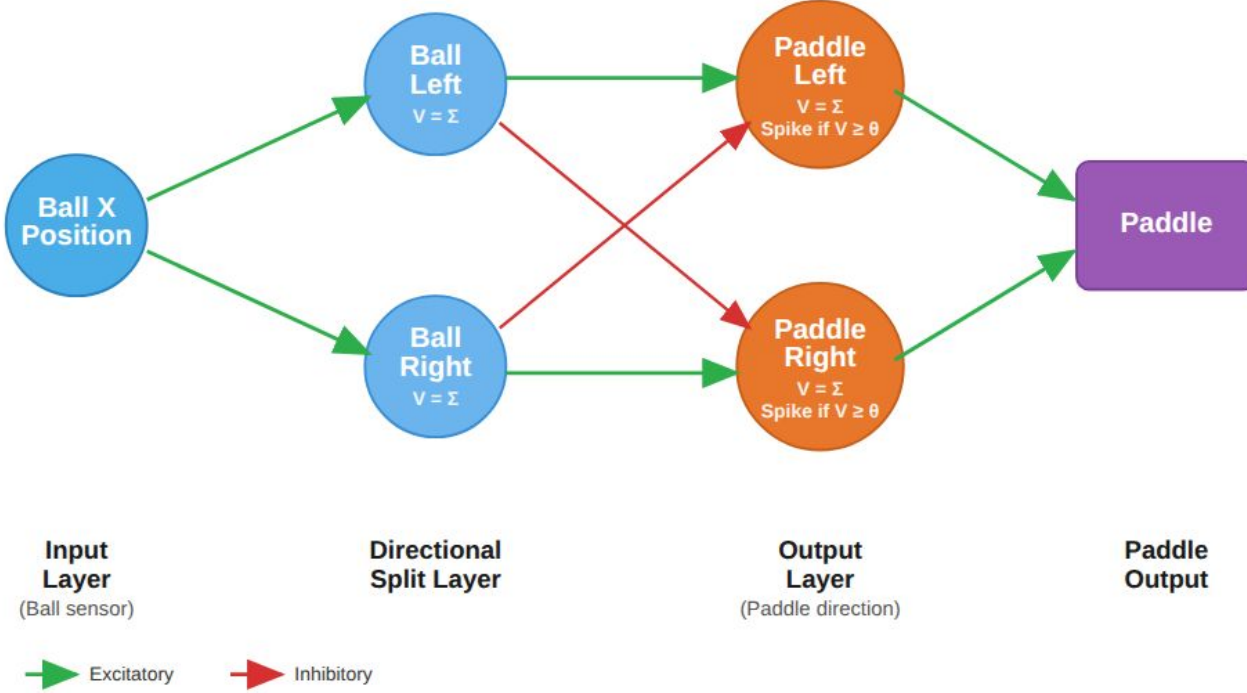


Figure 1 - Network Architecture Diagram

- System is split into two subsystems: an analog neuromorphic hardware SNN and a software game environment, bridged by an ESP32 microcontroller.
- The neuron circuits receive voltage signals from the game and fire after the charge hits a set threshold.
- The resulting voltage spikes pass through the synapse circuit which are read by the ESP32 and sent via universal asynchronous receiver-transmitter (UART) as game actions.

Methodology

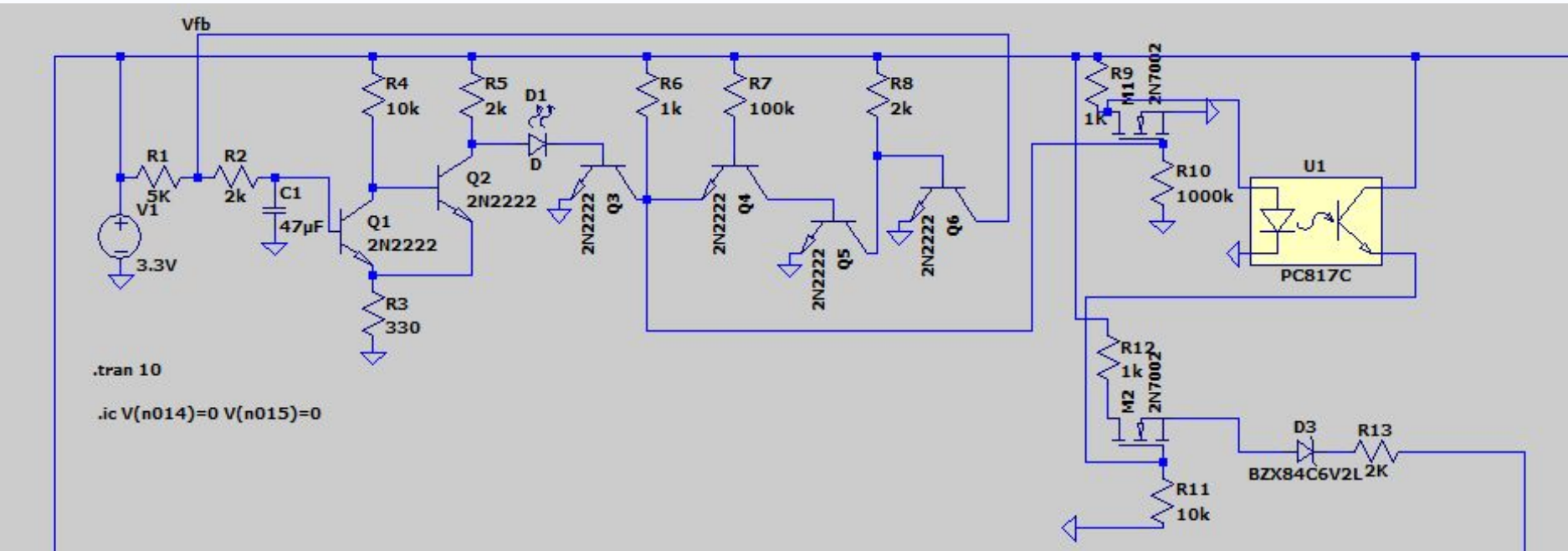


Figure 2 - Single neuron spice simulation

Initial steps included creating a single neuron simulation in LTspice (Figure 2) for design verification followed by simulating two more, an inhibitory and control neuron. From there, we validated the design with a physical breadboard neuron, proving we could replicate the expected spiking behavior before committing to a printed circuit board (PCB) layout.

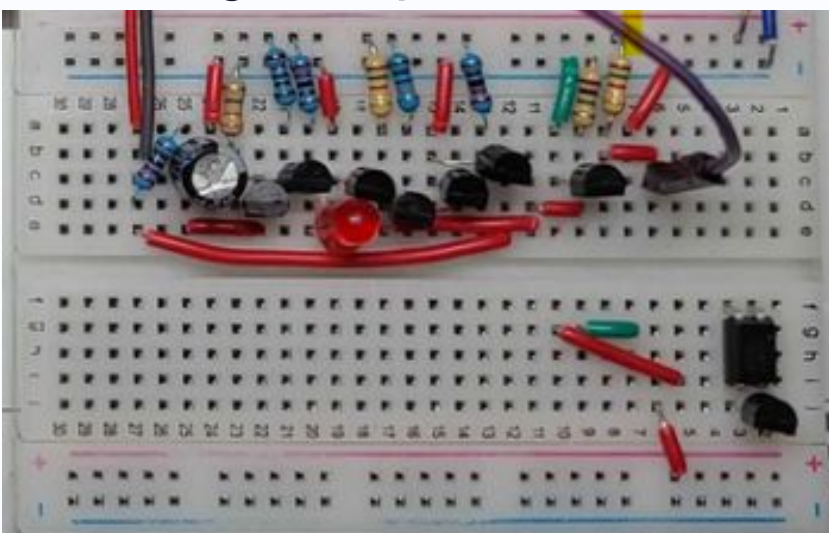


Figure 3 - Breadboard layout

Following the initial validation, we expanded the breadboard circuit to a 3-neural network. After hardware verification, breadboard schematic converted into custom PCB layout (Altium Designer). The project culminated in printing and physical assembly of our final PCB, translating the integrated and fire model into a robust, tileable device.

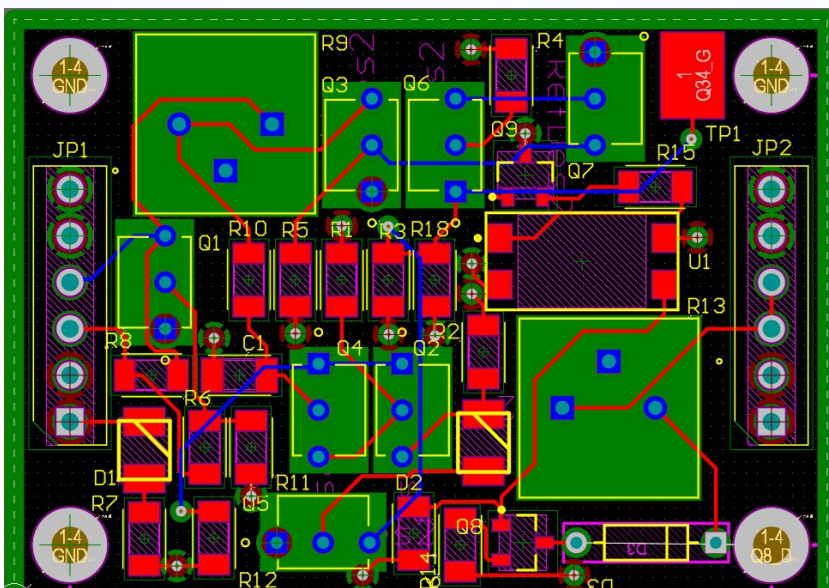


Figure 4 - PCB layout

Analysis and Results

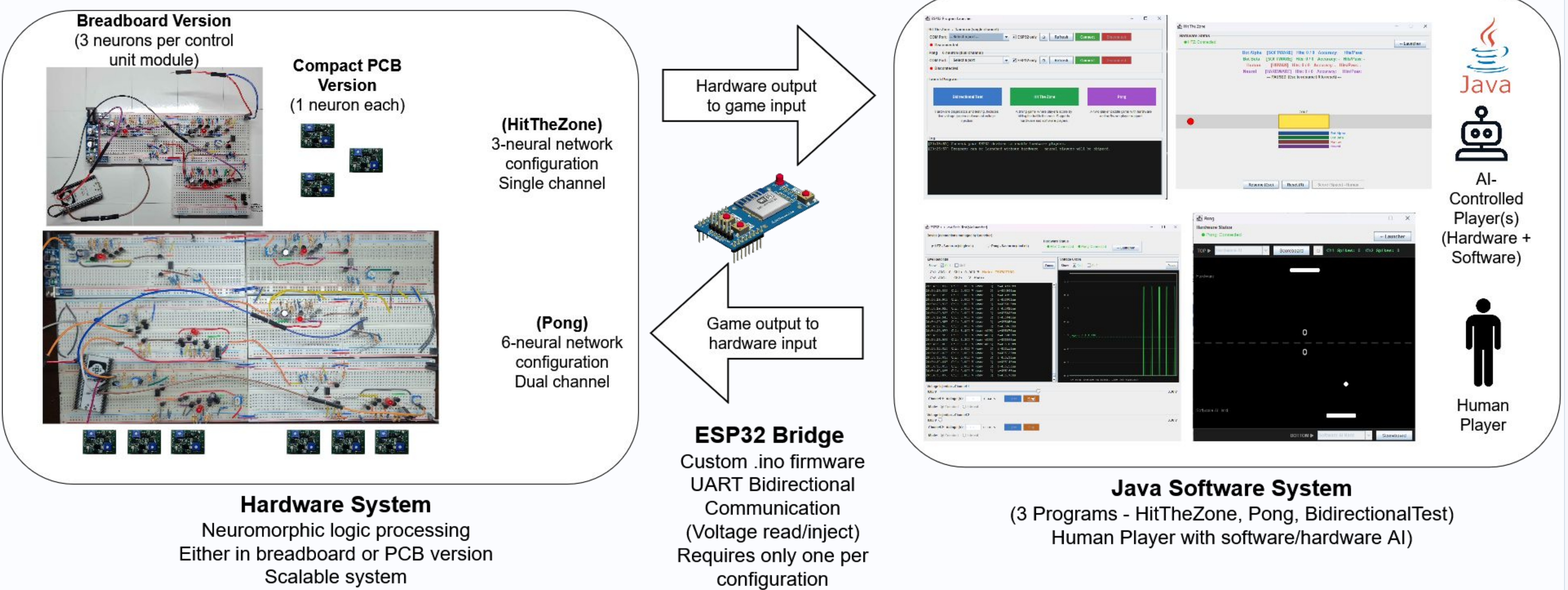


Figure 5 - Hardware and Software System High-Level Overview

Hardware

A basic control unit requires a 3-neural network: a control neuron, an inhibitory neuron, and excitatory neuron. A control unit represents one game action.

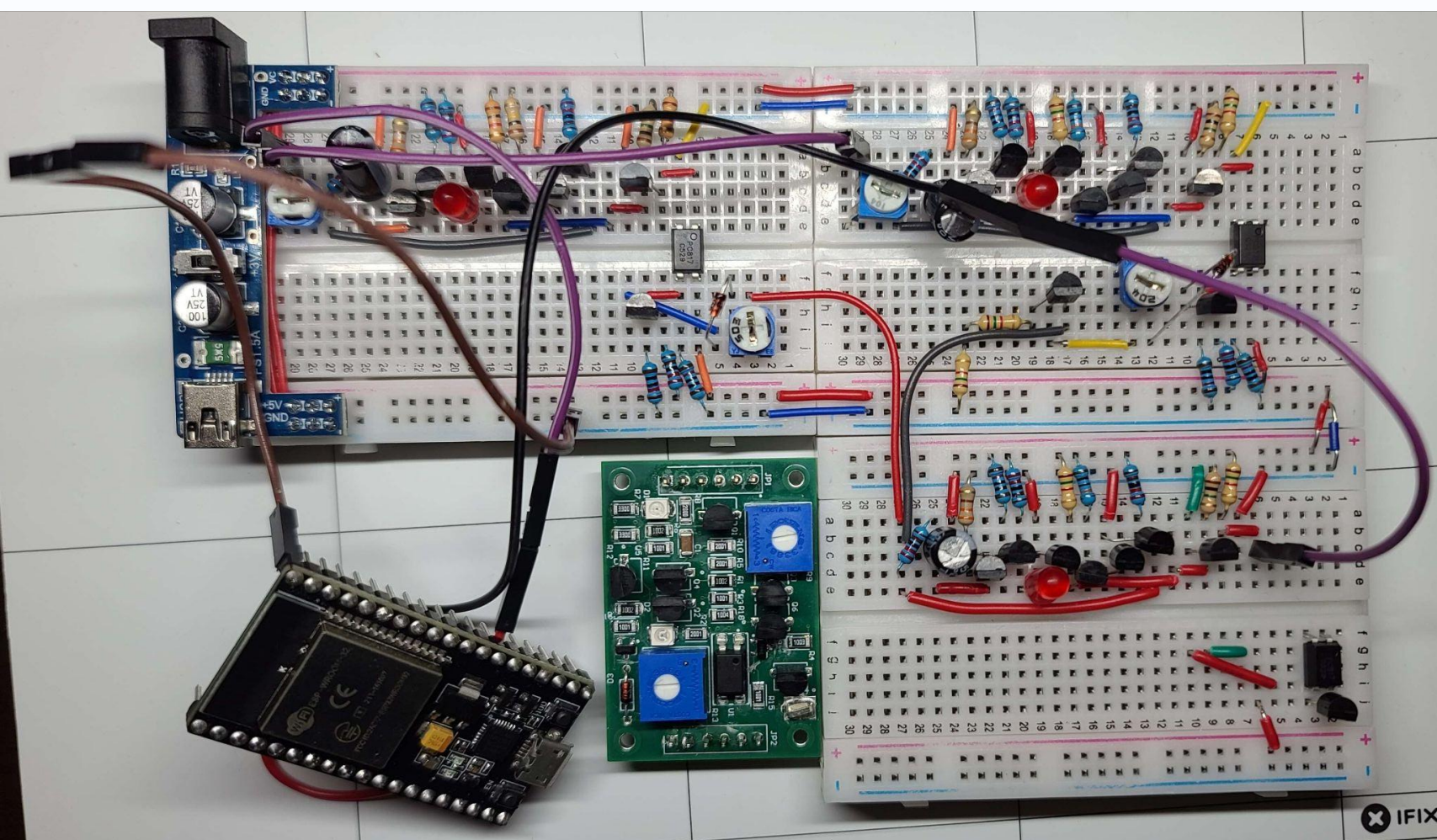


Figure 6 - Breadboard 3-neural network (HitTheZone)
For scale reference, one PCB neuron is shown

In Figure 6, a breadboard 3-neural network is shown:

- ESP32 injects stimulus into the excitatory neuron encouraging the control neuron to fire.
- Inhibitory neuron modulates the control neuron to suppress overfiring and improve signal stability of the control neuron.
- Read by the ESP32, the control neuron acts as the final integrated output of both the inhibitory and excitatory neurons.

This control unit scheme is scalable up to many game inputs. HitTheZone, a 3-neural network, supports one game input. Meanwhile, Pong is a 6-neuron network with two 3-neuron networks to supports up to two inputs. The breadboard and PCB neuron are functionally identical, but the PCB neuron is more compact and durable because of soldered on components compared to loosely mounted components on a breadboard. For demo purposes, the breadboard version better showcases individual components.

Software

Software includes ESP32 firmware and a software Java program with many subprograms.

- ESP32 .ino firmware (Single and dual channel)
- Java program Launcher
- Java game HitTheZone (3-neural network)
- Java game Pong (6-neural network)
- Java diagnostic tool BidirectionalTest

ESP32 firmware is written in Ardurino and delivers the stimulus to the excitatory neuron, read the control neuron's response and return output as real-time voltage. Java program is able to read discrete spikes and translate each spike into one discrete game event. In Figure 7, Pong game, controlled by a 6-neural network. Two control units (3-neural networks, Ch1 and Ch2) control left and right paddle movements respectively.

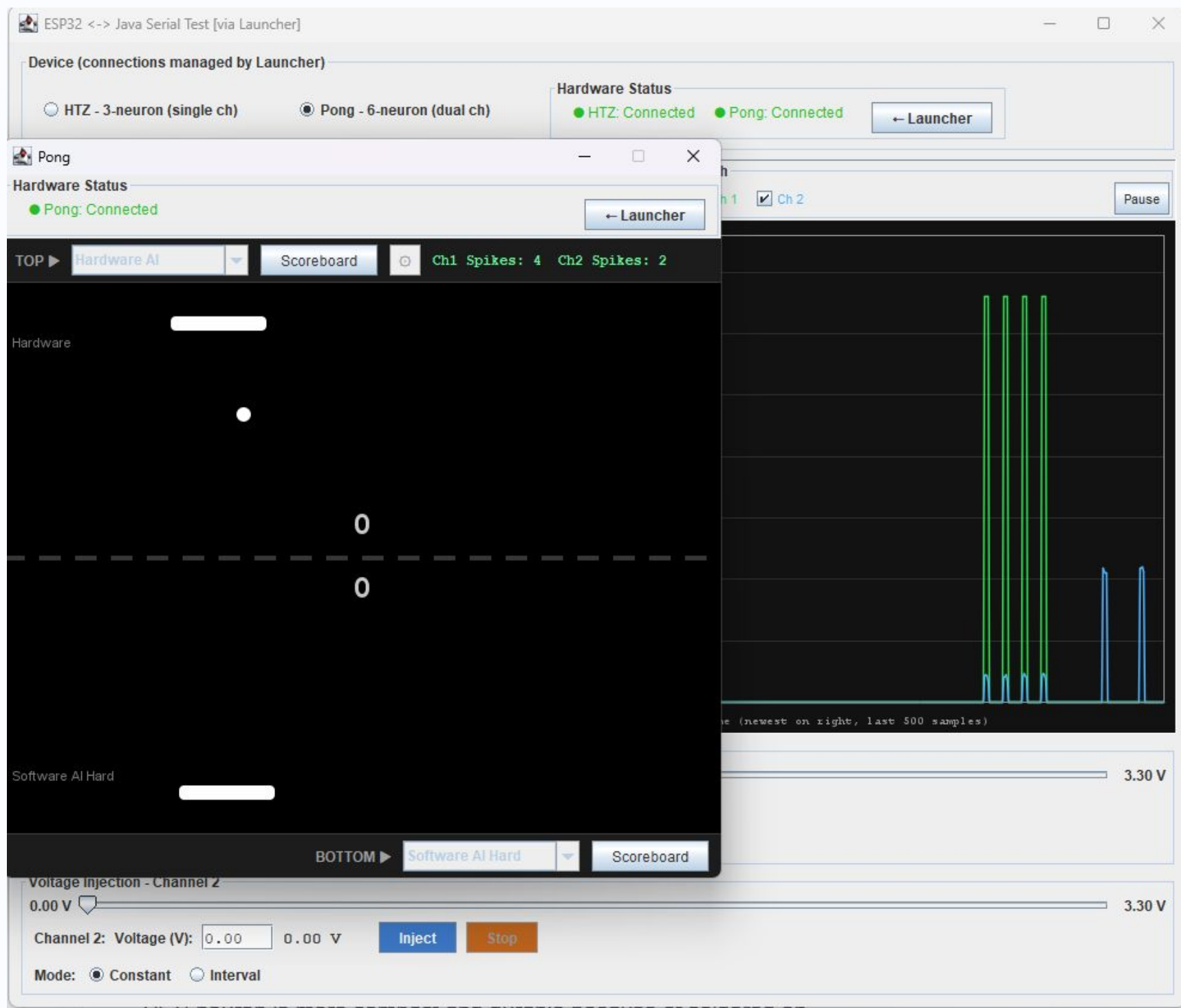


Figure 7 - Pong game overlaid over BidirectionalTestTool showing spike-based paddle control. Ch1 controls left and Ch2 controls right. One spike equals one game (paddle) event.

Hardware and Software Testing

- Software was validated using JUnit, Mockito, JaCoCo achieving 80% code coverage.
- Software correctly interprets spike events and are converted one-to-one with game actions.
- Hardware verification included a function generator, power supply, and oscilloscope confirming stable neuron firing, and tuning via potentiometers.
- Stress tests passed while handling repeated disconnect/reconnect events.

CI — Build & Test passing coverage 80%+

Summary/Conclusions

The project successfully demonstrated that a small-scale SNN built from analog components can control a system in real-time. The SNN tracked ball position and generated movement commands without any traditional processor or software-based interface involved in the decision-making. By bridging the neuron circuits with a Java game environment through ESP32, we showed that neuromorphic hardware can interface with conventional software systems using a simple microcontroller. The modular PCB layout allows for future expansion to larger and more complex designs.

Key References

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